Paper Summary

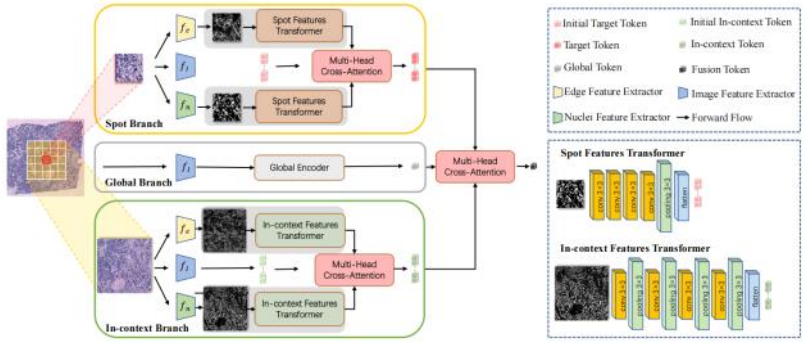
Wednesday, December 17, 2025 10:00 PM

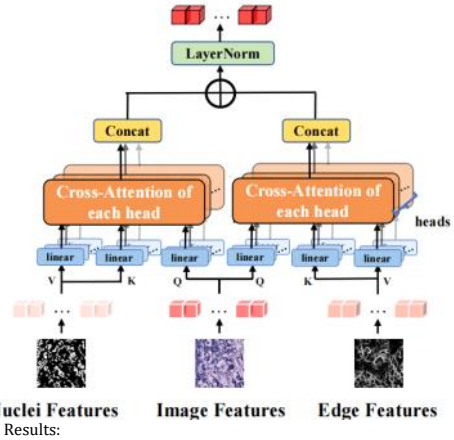
★ BG-TRIPLEX

Keywords:

- DL-based computational method
  - ST-Net
  - HisToGene
  - Hist2ST
  - EGGN
  - BLEEP
  - THItoGene
  - TRIPLEX
- Extracting boundary features models:
  - HED
  - DeepEdge
  - RCF
  - U-Net
  - MicroNet
  - SAMS-Net

- Resolved Problem
  - Fail to account for **cellular structure similarity, density, microenv interaction**

- Method:
  - 3 branches architecture:
    - Spot
    - In-context
    - Global
  - Leverage **BOUNDARY INFORMATION**: edge and nuclei characteristics as *guiding feature.*
  - **MULTI-HEAD CROSS-ATTENTION**: learning cellular morphology & microenv details
  - 

  - 

- Results:
  - **OUTPERFORMS EXISTING METHODS:** Calculating 250 most gene
  -

Models	HER2ST			STNet			Skin		
	MSE (↓)	PCC(M) (↑)	PCC(H) (↑)	MSE (↓)	PCC(M) (↑)	PCC(H) (↑)	MSE (↓)	PCC(M) (↑)	PCC(H) (↑)
ST-Net [13]	0.260 ± 0.04	0.194 ± 0.11	0.345 ± 0.16	0.209 ± 0.02	0.116 ± 0.06	0.223 ± 0.10	0.294 ± 0.07	0.274 ± 0.08	0.382 ± 0.08
EGN [16]	0.241 ± 0.06	0.197 ± 0.11	0.328 ± 0.17	0.192 ± 0.02	0.111 ± 0.05	0.203 ± 0.09	0.281 ± 0.08	0.281 ± 0.06	0.388 ± 0.06
BLEEP [17]	0.277 ± 0.05	0.151 ± 0.11	0.277 ± 0.10	0.235 ± 0.02	0.145 ± 0.05	0.193 ± 0.10	0.297 ± 0.08	0.269 ± 0.07	0.396 ± 0.08
HistoGene [14]	0.248 ± 0.05	0.178 ± 0.13	0.308 ± 0.14	0.199 ± 0.03	0.122 ± 0.10	0.201 ± 0.09	0.294 ± 0.07	0.312 ± 0.07	0.406 ± 0.10
His2ST [15]	0.285 ± 0.08	0.118 ± 0.10	0.258 ± 0.11	0.181 ± 0.07	0.144 ± 0.07	0.199 ± 0.07	0.291 ± 0.16	0.174 ± 0.16	0.353 ± 0.07
TRIPLEX [19]	**0.228± 0.07**	0.314 ± 0.14	0.497 ± 0.17	0.202 ± 0.07	0.206 ± 0.07	0.352 ± 0.10	0.268 ± 0.09	0.374 ± 0.07	0.490 ± 0.07
TG-TRIPLEX	0.277 ± 0.01	**0.366 ± 0.02**	**0.605 ± 0.02**	**0.1338 ± 0.01**	**0.4978 ± 0.02**	**0.7234 ± 0.04**	**0.128 ± 0.01**	**0.655 ± 0.01**	**0.752 ± 0.01**

★ DANet (Dynamic Alignment Network)

- Resolved Problems:
  - Extracting **fine-grained local features**
  - Modeling **long-range gene-gene correlations**
  - **Dynamic alignment** of image and genetic data in bimodal **contrastive learning**

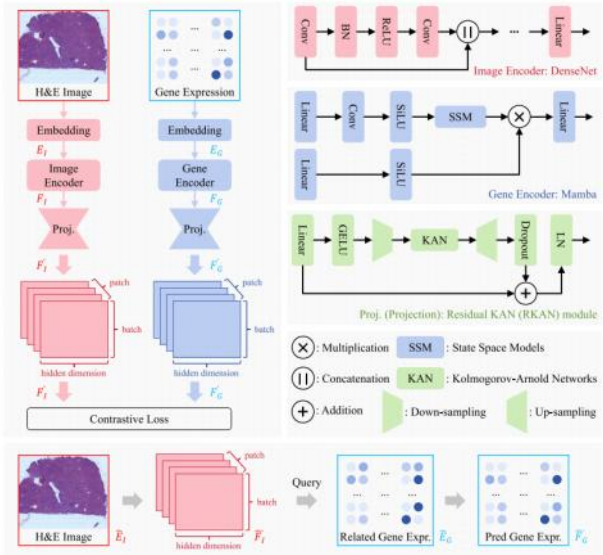
- Method:

IDEA:

- Bio-Image feature extraction models
- Ask NotebookLM to get the "Future development" part in paper that the author had pointed out

PROBLEMS:

- Fail to account for **cellular structure similarity, density, microenv interaction**
- Extracting **fine-grained local features**
- Modeling **long-range gene-gene correlations**
- **Dynamic alignment** of image and genetic data in bimodal **contrastive learning**
- **Genome-wide tumor mRNA expression** thus generate imputed transcriptomics that serve as a general, WSI-based predictor of patient response
- To overcome the limitations of **spot-based spatial transcriptomics** (which capture cell mixtures) by **predicting gene expression at the finer, single-cell level**
- **Adaptive hierarchical clustering** in single-cell RNA sequencing
- Capture complex hierarchical structures of cell types and subtypes by **minimizing topological entropy (TE)**
- **Hierarchical outputs** reflect the developmental potency and intra-diversity of cell populations

- Feature Extraction: **DenseNet** - reuse and capture local refinement
- Gene-gene Correlation: **State Space Models (SSMs)** - Mamba module in Gene-Encoder
- Dynamic Alignment: **Residual Kolmogorov-Arnold Network (RKAN)** - adjust bimodal mappings based on input characteristics
- 

- Performance:
  - Dataset: **GSE240429 (5 genes)**
  -

HisToGene		ST-Net		BLEEP		DANet	
Gene Name	**PCC**	**Gene Name**	**PCC**	**Gene Name**	**PCC**	**Gene Name**	**PCC**
CYP3A4	0.549	CYP3A4	0.549	CYP3A4	**0.741**	CYP3A4	0.739
CYP1A2	0.542	CYP1A2	0.532	CYP1A2	**0.681**	ORM1	**0.681**
GLUL	0.488	CYP2E1	0.530	CYP2E1	**0.675**	CYP2E1	0.673
CYP2E1	0.330	GLUL	0.463	GLUL	0.656	CYP1A2	**0.661**
FABP1	0.328	SLCO1B3	0.375	FABP1	0.503	GLUL	**0.624**
Avg.	0.447	Avg.	0.490	Avg.	0.651	Avg.	**0.676**

  - Dataset: **HER2ST (5 genes)**
  -

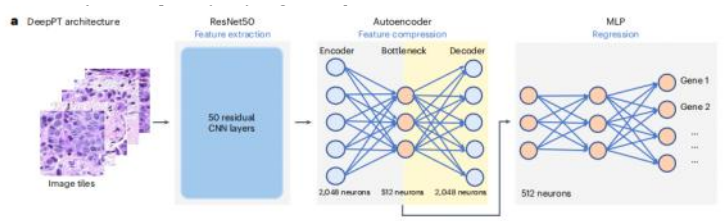
	Method	Gene	PCC	Gene	PCC	Gene	PCC	Gene	PCC	Gene	PCC	Avg.
A1	HisToGene*	FASN	0.378	GNAS	0.275	SCD	0.271	DMAP1	0.188	AKR1A1	0.180	0.258
	ST-Net*	GATC	0.192	EZH2	0.176	FLNA	0.163	CCT4	0.162	FHF11	0.160	0.171
	BLEEP	Gene	0.422	SCD	0.366	MUCL1	0.342	CXCL10	0.320	IGKC	0.287	**0.347**
	DANet	FASN	0.355	SCD	0.332	C3	0.278	GNAS	0.270	MUCL1	0.218	0.291
B1	HisToGene*	GNAS	0.532	FN1	0.498	RHOB	0.496	GBP3	0.471	TMEM123	0.469	0.493
	ST-Net*	TMEM14B	0.484	FN1	0.440	GNAS	0.411	SCD	0.393	CPNE1	0.386	0.423
	BLEEP	TMEM14B	0.560	FN1	0.553	FAM117A	0.532	SDHB	0.512	COL4A1	0.507	0.533
	DANet	FN1	0.584	TMEM14B	0.576	FAM117A	0.563	STMN1	0.537	ZFP36	0.537	**0.559**
C1	HisToGene*	GNAS	0.430	CLDN4	0.391	NDUFB9	0.376	FDPS	0.346	PXDN	0.345	0.378
	ST-Net*	SAT1	0.356	CPSF1	0.269	TMEM14B	0.256	GNAS	0.255	TMEM123	0.238	0.275
	BLEEP	UFD1L	0.471	FDPS	0.470	GNAS	0.458	ZDHHC12	0.453	CLDN4	0.453	**0.461**
	DANet	ITGB6	0.441	GNAS	0.435	UFD1L	0.402	ZDHHC12	0.383	MRPL51	0.377	0.408
D1	HisToGene*	FN1	0.517	SCD	0.453	ITGB6	0.436	C3	0.392	SRSF1	0.389	0.437
	ST-Net*	TMEM123	0.242	SCD	0.228	FN1	0.223	BTG1	0.207	PPP1R37	0.201	0.220
	BLEEP	FN1	0.466	SCD	0.432	ITGB6	0.398	C3	0.397	SRSF1	0.397	0.418
	DANet	C3	0.523	FN1	0.518	SCD	0.495	ITGB6	0.469	TXNDC17	0.444	**0.490**
E1	HisToGene*	IGHG3	0.132	IGHM	0.129	IFI27	0.125	DNAJB2	0.114	PLXND1	0.109	0.122
	ST-Net*	SLC35A2	0.161	JCHAIN	0.137	PLVAP	0.110	CGGBP1	0.109	IGLC7	0.108	0.125
	BLEEP	IGKC	0.141	IGHG4	0.126	CD14	0.119	IGLC3	0.116	IFI27	0.114	0.123
	DANet	IGKC	0.160	IGLC2	0.153	S100A9	0.150	TMEM86A	0.122	ANKLE2	0.122	**0.141**
F1	HisToGene*	FASN	0.190	SCD	0.183	LUC7L3	0.178	MYL9	0.154	GNAS	0.154	0.172
	ST-Net*	GNAS	0.147	LIMCH1	0.140	JAK3	0.110	IGLC7	0.108	PLP2	0.107	0.122
	BLEEP	LUC7L3	0.173	SCD	0.149	GNAS	0.141	HCST	0.141	IGKC	0.131	0.147
	DANet	IGKC	0.341	IGLC2	0.289	FASN	0.259	IGHG3	0.233	IGLC3	0.228	**0.270**
G2	HisToGene*	TMEM123	0.404	GNAS	0.383	NDRG1	0.371	CLDN4	0.369	FN1	0.360	0.377
	ST-Net*	CLDN4	0.302	TMEM123	0.281	BTG1	0.234	SAT1	0.225	MIF	0.219	0.252
	BLEEP	TMEM123	0.399	GNAS	0.391	ITGB6	0.390	MYL12B	0.382	ZNF217	0.372	**0.387**
	DANet	GNAS	0.379	TMEM123	0.350	CX3CL1	0.348	FASN	0.337	ITGB6	0.330	0.349
H1	HisToGene*	GNAS	0.426	HLA-DRA	0.425	ITGB6	0.376	FASN	0.351	IGHA1	0.340	0.384
	ST-Net*	HLA-DRA	0.314	CPNE1	0.270	BTG1	0.252	SRSF5	0.247	GATA3	0.232	0.263
	BLEEP	IGKC	0.485	GNAS	0.439	IGHG4	0.417	HLA-DRA	0.406	MGP	0.391	0.428
	DANet	IGKC	0.481	GNAS	0.453	IGHG4	0.439	HLA-DRA	0.429	ITGB6	0.421	**0.445**

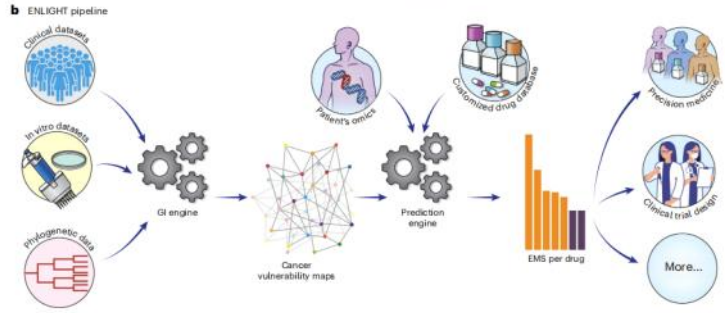
★ DeepPT (Deep Pathology for Transcriptomics)

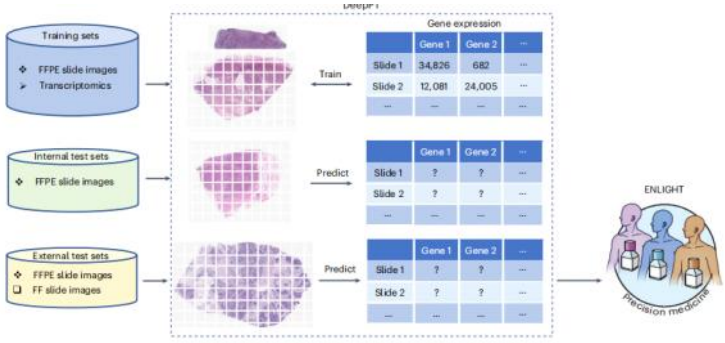
Inside larger indirect: **ENLIGHT-DeepPT**: 2 step approaches

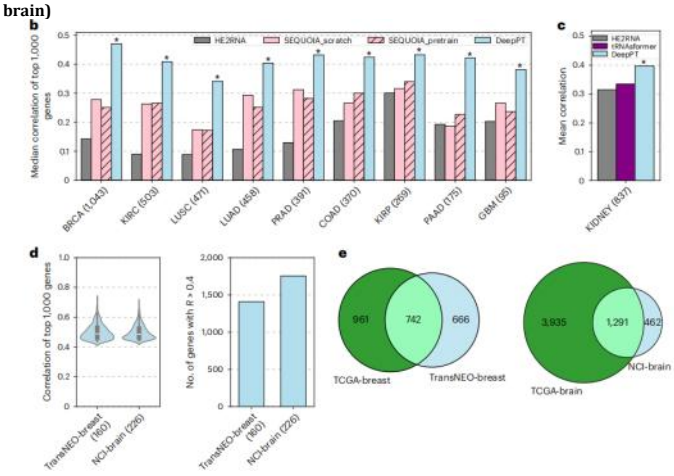
- Objective:
  - **Genome-wide tumor mRNA expression** thus generate imputed transcriptomics that serve as a general, WSI-based predictor of patient response

- Method:
  - 3 main units:
    - Feature extration unit: pretrained **ResNet50**
    - Feature compression unit: **autoencoder** 512 neuron-bottleneck: reduce noise and overfitting
    - **MultiLayer Perceptron (MLP)**: Regression predict

- 

- 

- 

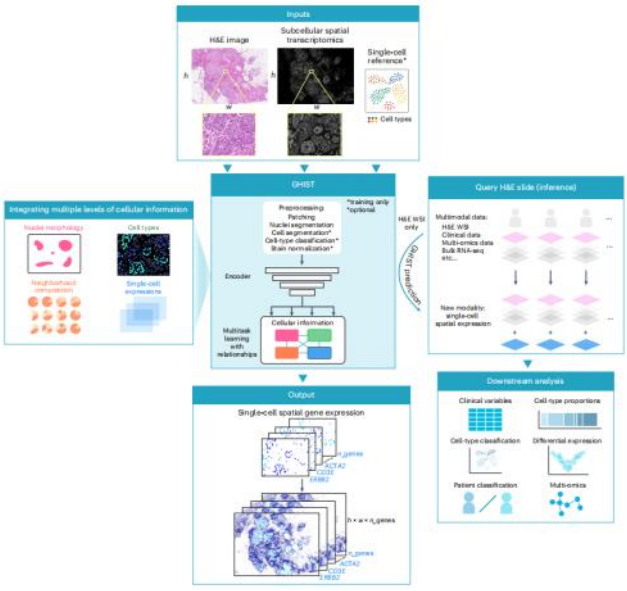
- Performance:
  - Dataset: **16 The Cancer Genome Atlas (TCGA) cohort** & **generalized well to two independent external datasets (TransNEO-breast and NCI-brain)**
  - 

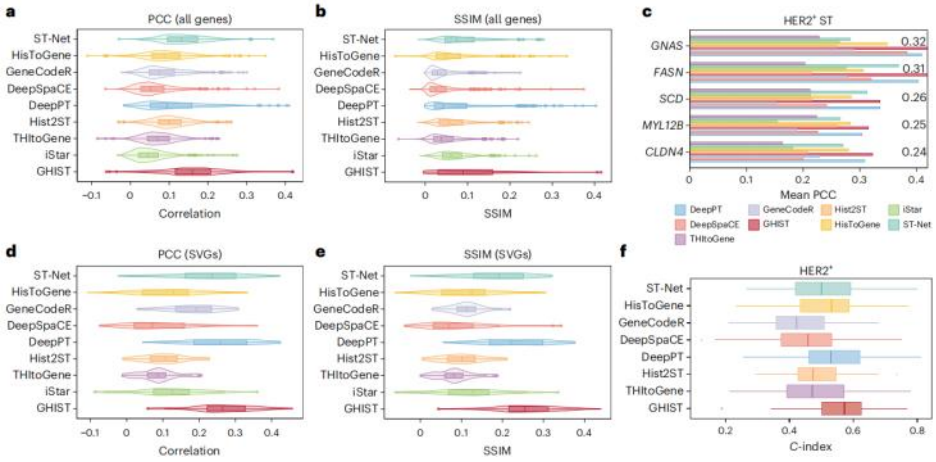
★ GHIST (Gene expression from HISTology)

- Objective
  - Multi-task DL to predict **spatially resolved gene expression at single-cell resolution**

- Resolved Problems
  - To overcome the limitations of **spot-based spatial transcriptomics** (which capture cell mixtures) by **predicting gene expression at the finer, single-cell level**

- Method:
  - **4 prediction heads:**
    - Cell type
    - Neighborhood composition
    - Cell nucleus morphology
    - Single-cell RNA expression
  - Auxiliary task

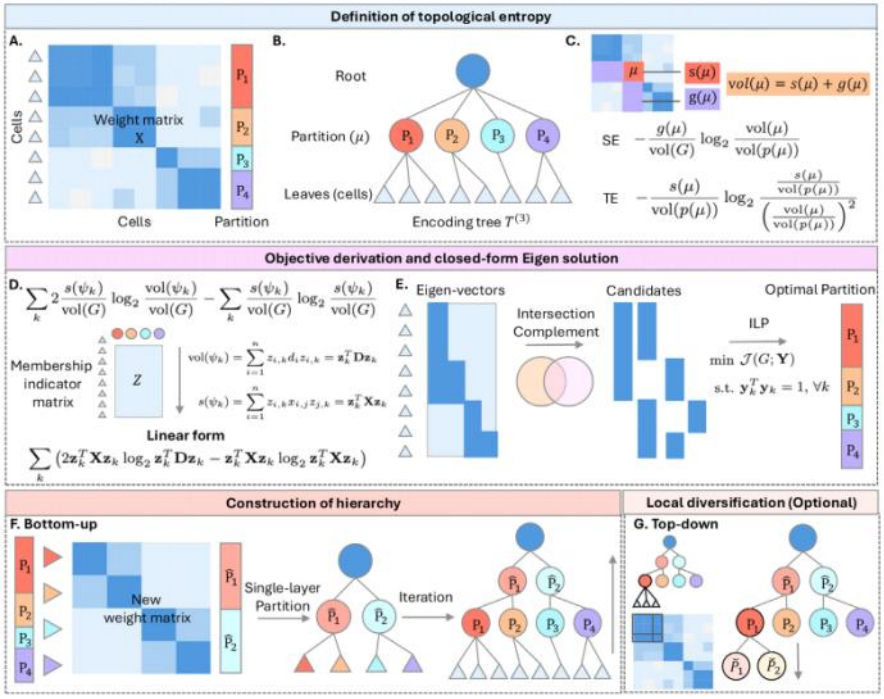
- 

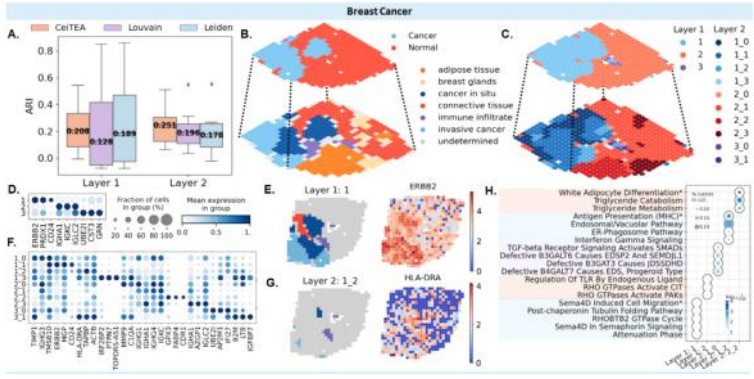
- Performance:
  - Dataset: **HER2ST** (Full gene)
  - 

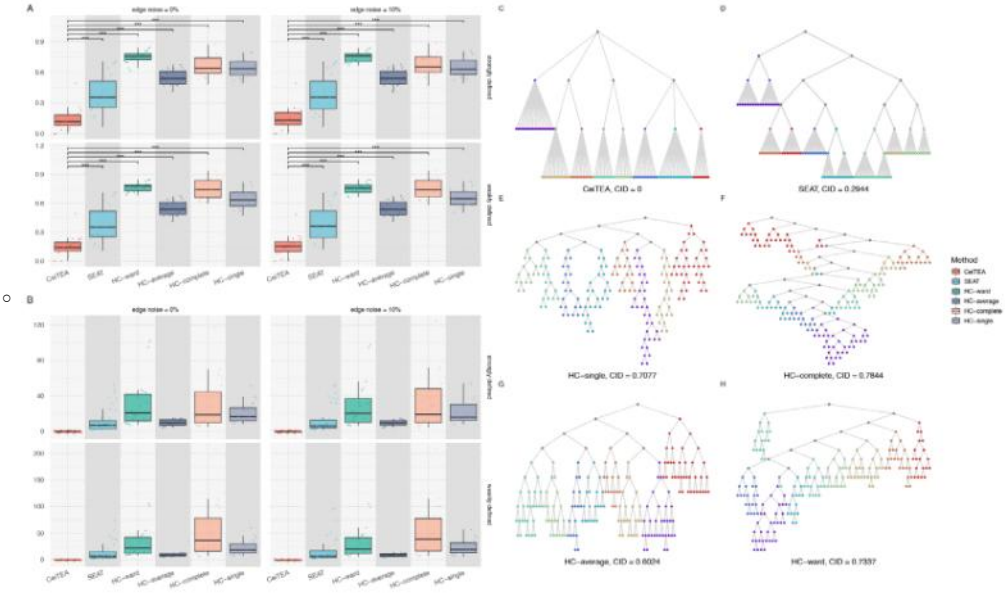
★ CeiTEA (Adaptive Hierarchy of Single Cells with Topological Entropy)

- Objective:
  - **Adaptive hierarchical clustering** in single-cell RNA sequencing
  - Capture complex hierarchical structures of cell types and subtypes by **minimizing topological entropy (TE)**
  - **Hierarchical outputs** reflect the developmental potency and intra-diversity of cell populations

- Method:
  - Construct a root, unbalanced **multi-nary partition tree** w/o rigid structural constraints, use of **Topological Entropy (TE)** emphasize local connectivity and incorporates the hierarchical structure of the partition tree.
  - 2 Steps strategy:
    - **Bottom-up construction:** generates a minimal TE single-layer partition using **Eigen decomposition** and linear programming
    - **Top-down Diversification (optional):** resonstruct subtrees to refine local diversity

- 

- Performance
  - 

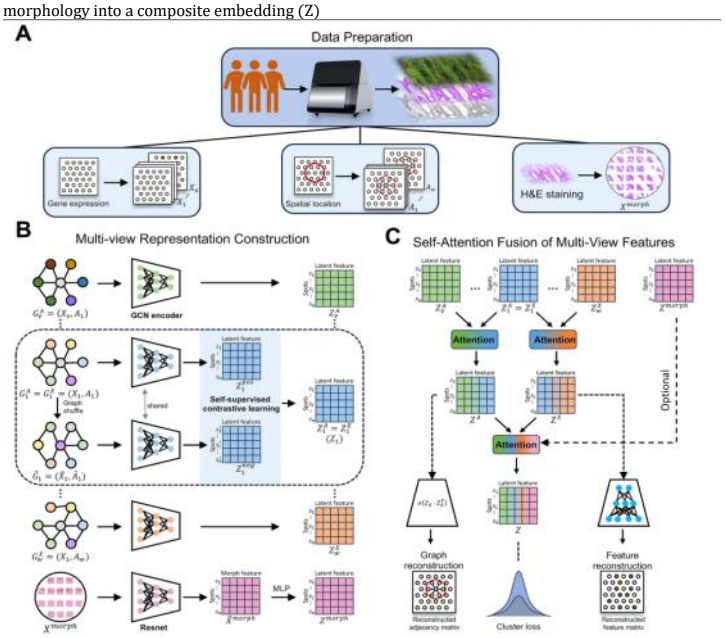
  - 

★ MCGAE (Multi-view Constrastive Graph Autoencoder)

- Objective:
  - **Accurate spatial domain identification:** in ST data by tackling challenges related to integrating multimodal data:
    - Gene Expression
    - Spatial Coordinates
    - Morphological Information

- Aim to: generate a multimodal, multi-view biological **representation** that is robust to noise and cariability

- Architecture and Mechanism:
  - Multi-view representations using a **modular modeling approach** applied to gene expression (X) and spatiadjacency matrices (A). Use **Contrastive graph convolutional networks (GCNs)** coupled with attention modules to learn modality-specific spot representations
  1. **Multi-view Generation:** Generate multiple topological graphs from spatial coordinates and uses GCNs to integrate gene expression and spatial coordinates, which is further refined by **self-supervised contrastive learning** to capture genuine biological signals
  2. **Image Integration:**
     If morphological images are available, a pre-trained **ResNet50** extracts image features, mapping them into a spot-like representation space (Z_{morph})
  3. **Fusion:** A final **attention mechanism** combines the fused embeddings related to gene expression, adjacency matrix reconstruction, and image morphology into a composite embedding (Z)

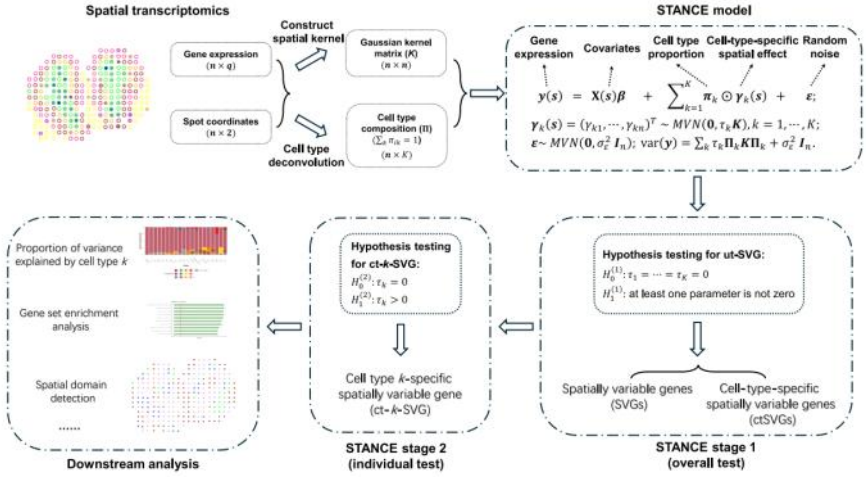


- Performance/Contribution:
  - Dataset: **DLPFC, HBC**
  - **Superior performance in spatial domain detection** on diverse ST dataset
  - Metrics: **Adjusted Rand Index (ARI)**
  - Capable of **data denoising**
  - **Tumor invasion regions**

★ STANCE (Spatial Transcriptomics Analysis of genes with Cell-type-specific Expression)

- Objective:
  - Unified **statistical model** developed to detect both **spatially variable genes (SVGs)** & **cell-type-specific spatially variable genes (ctSVGs)**.
  - Ensure detection results are **invariant to spatial rotation** - which is a critical issue in existing fixed-effects models.

- Architecture and Mechanism:
  - Utilizes: **Linear Mixed-effect model framework** integrate gene expression, spatial location and cell type composition
  1. **Rotation Invariance:** It achieves spatial rotation invariance by constructing a kernel matrix (K) based solely on the **Euclidean distance** between spatial spots, ensuring the matrix is unaffected by rotations
  2. **Spatial Decomposition:** The spatial random effects (γ(s)) are decomposed into K cell-type-specific components, weighted by the cell type composition (Π)
  3. **Two-Stage Testing:**
     Stage 1 performs an overall test to identify unified-type SVGs (utSVGs), which include both SVGs and ctSVGs. Stage 2 conducts an individual test on utSVGs to identify specific ctSVGs by testing whether the corresponding cell-type spatial variance component (τ_k) is non-zero

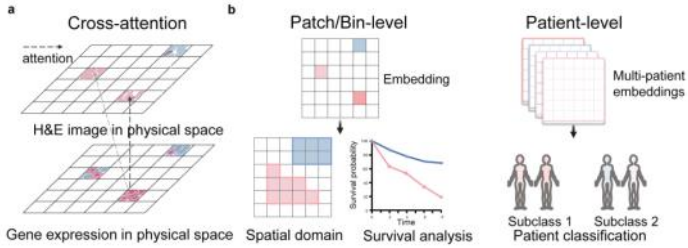


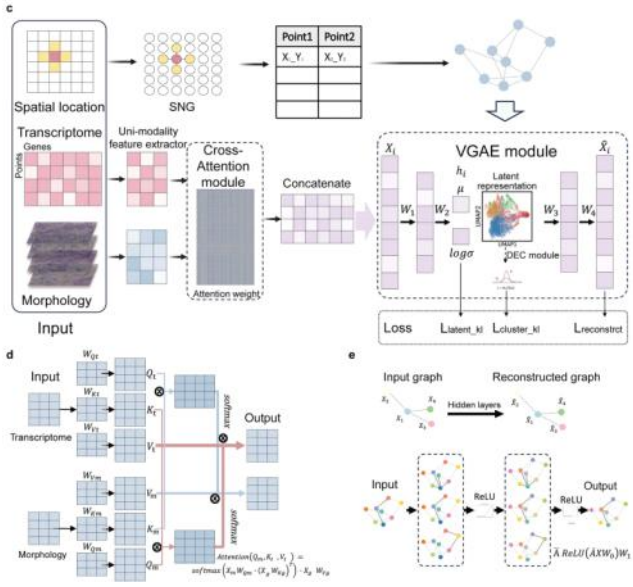
- Performance/Contribution:
  - MCGAE demonstrated **superior performance in spatial domain detection** on diverse ST datasets. Integrating image features with MCGAE significantly increased the Adjusted Rand Index (ARI) score (e.g., from 0.53 to 0.64 in Human Breast Cancer data). It is capa ble of **data denoising**

(resulting in significantly higher spatial correlation, 0.76, for spatially variable genes (SVGs) derived from denoised data compared to raw data),, and effectively facilitates the study of **tumor invasion regions** in colorectal cancer liver metastasis

★ StereoMM

- Objective:
  - The integration of multimodal spatial omics data:
    - Gene Expression
    - Histological images
    - Spatial location
  - Facilitate comprehensive biological and clinic discoveries
  - Aim to: achieve holistic patient characterization and can be used for **patient-level classification**.

- Architecture and Mechanism:
  - **Cross-Attention Module:** This module facilitates **inter-modal interaction** between gene expression and morphological features. It dynamically allocates weights across the entire tissue slice via the cross-attention mechanism to capture **long-range dependencies**, rather than just local neighbors
  - **Variational Graph Autoencoder (VGAE):** This module aggregates spatial neighbor information using a **Spatial Neighbour Graph (SNG)** and the concatenated multimodal features,. The encoder utilizes GNNs (defaulting to GCN) to learn the latent representation (Z) that is sampled for graph reconstruction via the decoder
  - 

  - 

- Performance/Contribution:
  - StereoMM achieved **superior performance** in spatial domain recognition compared to single modalities and competing algorithms on human lung adenosquamous carcinoma and mouse brain data,. It effectively revealed **tumor heterogeneity** in breast cancer, identifying distinct tumor regions with varying Copy Number Variation (CNV) and stemness scores. Furthermore, it successfully enabled accurate **patient classification** (e.g., distinguishing dMMR/pMMR colorectal cancer patients) using a graph representation constructed from its derived spatial domain s